# KRISHNA MULA

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## EXPERIENCE

**Intake AI**
Software Engineer Intern — Jun 2026 – Present

- Led the migration of production off an unmanaged EC2 host onto **isolated dev and prod AWS accounts** (ECS Fargate, SQS), provisioned in **Terraform** with **CI/CD** via branch-locked GitHub OIDC deploys.
- Spearheaded the live migration of a shared MongoDB database into **isolated per-customer databases** via **freeze-copy-verify**, moving **103,872 records across 9 production organizations** with **no platform downtime**.
- Architected the platform-wide telemetry **event store**: a transactional outbox streaming every service's events to **S3 and ClickHouse**, Parquet compaction that cut storage **36×**, and **Grafana dashboards** that surface agent failures.
- Proposed, built, and shipped a **GPU reranking service** (cross-encoder, FP16 on an NVIDIA L4) for a clinical-trial **RAG assistant**, raising how often the correct passage lands in the top 5 results from **64% to 78%**.
- Built a **GPU embedding service** for the RAG assistant, serving its model on a shared GPU behind a **priority queue**, cutting query embedding from **∼1 s to 17 ms** and a worst-case chat from **8 minutes to 16 seconds**.
- Built an **automated GPU profiling harness** that tuned the reranking service's batch size from 32 to 4, cutting the **p50 latency of chat responses by over 300 ms** (reranking 986 to 662 ms) and **peak GPU memory by 46%**.

**PwC**
Software Engineer — Aug 2022 – Jul 2025

- Architected a **RAG pipeline** for IT security compliance, embedding documents into ChromaDB for semantic retrieval and integrating **LLM inference** for automated validation against regulatory standards, **reducing review time by 70%**.
- Designed a **RabbitMQ-backed async ingestion architecture** that decoupled upload APIs from parsing, chunking, embedding, and vector indexing, **reducing latency by 30%** under bursty workloads.
- Built an ETL pipeline using PyRFC to extract SM20/SM21 security audit logs from **5 production SAP systems**, replacing a process that required **2+ hours of manual export and formatting** per report.
- Automated security controls execution on client SAP systems by codifying **25+ controls** as Flask + MongoDB REST APIs, ingesting data directly from SAP HANA and validating against security specifications.
- Built **serverless compliance reporting APIs on AWS Lambda** for 15+ client systems, pre-warmed to handle twice-daily traffic bursts without cold-start delays.

Software Engineering Intern — Feb 2022 – Aug 2022

- Built Nivo.js dashboards tracking security control pass/fail rates, compliance coverage, and historical trends, letting teams prioritize which controls to automate next.
- Automated deployments with **Azure CI/CD** pipelines, reducing release cycles from **2 days to 4.5 hours**.

## PROJECTS

**WarpCache**: **CUDA decode-attention kernel** over a paged **INT8/FP16 KV cache**; **Flash-Decoding split-KV** cut batch-1 decode at a 16K-token history **50×** (8.1 ms to 162 $\mu$s) on an RTX 3090. **Nsight Compute** profiling showed INT8 halves cache memory but not latency, as the kernel is occupancy-bound; static K smoothing cut **quantization error** from **6.1% to 1.8%**.

**Aether**: Distributed **pub-sub system over raw TCP** with a gossip-based broker mesh and **Chandy-Lamport snapshots** replicated to k peers for failover; subscribers detect failure and self-heal without coordinated restarts.

## TECHNICAL SKILLS

**Languages**: Python, TypeScript, SQL, C++, CUDA

**ML & Inference**: PyTorch, Hugging Face Transformers, cross-encoder reranking, fp16 GPU serving, ChromaDB

**Frameworks**: FastAPI, Flask, Node.js, React, Next.js

**Infrastructure**: Terraform, Docker, AWS (ECS, Fargate, EC2, ECR, Lambda, SQS, S3, RDS, DynamoDB, CloudWatch, IAM/OIDC), Azure (Functions, VM, Entra ID), GitHub Actions, Nginx

**Data & Messaging**: PostgreSQL, MongoDB, ClickHouse, Redis, Cassandra, SQLite, RabbitMQ, SQS

**Observability**: OpenTelemetry, Prometheus, Grafana, Loki, Sentry

## EDUCATION

**Master of Science in Computer Science**, San Jose State University — Aug 2025 – May 2027
*Coursework:* Distributed Systems, Advanced Parallel Processing, Cloud Computing, Database Systems, Machine Learning